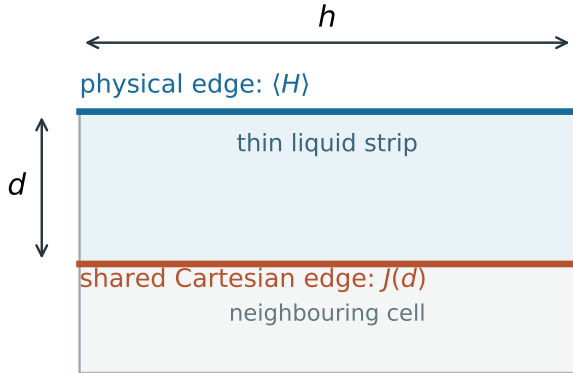


One trace, shared across two edge moments



Source-derived boundary data

$$J(d) = \langle H \rangle - d\langle N \rangle + \frac{d^2}{2}\langle S \rangle$$

$$\bar{u}_J = \frac{\langle H \rangle - J(d)}{d}$$

$$= \langle N \rangle - \frac{d}{2}\langle S \rangle$$

The same trace average cancels.

Optional, separate receiver assignment

$$\mathbf{v}_K^{\text{mix}} = \frac{\rho_l q_K \bar{\mathbf{u}}_{l,K} + \rho_g (c_{s,K} - q_K) \bar{\mathbf{u}}_{g,K}}{\rho_l q_K + \rho_g (c_{s,K} - q_K)}$$

Applied after liquid phase-volume moments are formed; it is not part of their closure identity.